Palak Deb Patra

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EDUCATION

National Institute of Technology, Rourkela **Dec 2021 – Jun 2025**
Bachelor of Technology in **Computer Science & Engineering** | **CGPA: 8.52/10** **Rourkela, India**

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming, Machine Learning, Natural Language Processing, Computer Vision, Deep Learning, Computer Networks.

PROFESSIONAL EXPERIENCE

NatWest Group | **Software Engineer** **Jul 2025 – Present**
Python, FastMCP, Azure, Microsoft Copilot Studio, Power Platform **Bangalore, India**

- Developed components of a **10-tool MCP server** for an AI learning assistant in Microsoft Teams, connecting Workday, ECDF, GitLab, Outlook, Confluence, and Jira.
- Implemented AI tracing with **OpenTelemetry** and **Azure Application Insights**, covering tool calls, conversations, and **8 failure categories**. Added **131 automated tests** and monitoring dashboards.
- Built a GitLab metrics pipeline using **Azure Blob Storage** and change-aware parsing, and connected Workday learning history to provide personalized insights.
- Created **3 interactive Microsoft Teams dashboards** and reduced their footprint by **35%**.
- Used **SAML/Entra ID** claims and a **3-stage fallback** to resolve employee identities securely for personalized responses.
- Created a **29-prompt regression suite**, increasing validation pass rates from **86% to 97%**. Automated deployments across **4 environments** using GitLab CI/CD and Ansible Automation Platform.

NatWest Group | **Software Engineering Intern** **May 2024 – Jul 2024**
Python, GraphQL, REST APIs, Snowflake, ThoughtSpot **Gurgaon, India**

- Collected organization-wide DevOps metrics through GitLab GraphQL and REST APIs. Automated paginated data extraction and CSV exports, reducing processing time by **28%**.
- Loaded **16 datasets** into Snowflake and built a shared data model for interactive ThoughtSpot dashboards.

PERSONAL PROJECTS

CodeShift | Python, LangGraph, Ollama, tree-sitter, Docker, TypeScript | GitHub **Aug 2026**

- Built a **six-agent system** with LangGraph to migrate Python projects to TypeScript, processing modules in dependency order with a locally hosted language model.
- Checked translated modules against the originals in Docker-isolated tests by comparing outputs, exceptions, and object state. Used **mypy** and **tsc** for static checks; **190 tests passed**.

Memex | Python, LangGraph, ChromaDB, Ollama, Streamlit | GitHub **Aug 2026**

- Developed an **offline retrieval-augmented generation system** that indexed **3,050 files** into **42,743 chunks** in under **15 minutes** and returned answers with source citations.
- Combined dense retrieval and BM25 using **Reciprocal Rank Fusion** and a cross-encoder reranker. Evaluated retrieval with **Hit@K, MRR, and Recall@5**.

TECHNICAL SKILLS

Languages	Python, Java, C++, JavaScript, TypeScript, SQL
AI & ML	PyTorch, TensorFlow, Scikit-learn, OpenCV, spaCy, NLTK, NVIDIA CUDA
Agents	LangGraph, LangChain, Huggingface, Ollama, FastMCP, Claude, Codex
Web & APIs	React, Angular, Streamlit, FastAPI, Spring Boot, REST APIs, GraphQL
Data & Cloud	Snowflake, Oracle, MySQL, ChromaDB, AWS, Azure
Tools	Git, GitLab CI/CD, Ansible, Docker, Jupyter

ACHIEVEMENTS

- Selected for the **Amazon ML Summer School 2023**, collaborating with Amazon Scientists on advanced ML topics.
- Solved **800+** LeetCode problems with a peak rating of **1795**, ranking in the **top 8%** globally.
- Achieved global ranks of **277** in Weekly Contest 438 and **1839** in Weekly Contest 437 on LeetCode.